Automated Applicant Screening for Resume Filtering Using Artificial Intelligence

Business Process 1:

User Stories

US1: Upload Resume
As a candidate, I want to access the company portal and upload my resume so that I can apply for a job.

US2: Receive Confirmation Email
As a candidate, I want to receive a confirmation email after submitting my application so that I know it was successfully received.

US3: Validate Resume Format
As an AI system, I want to validate the uploaded resume format (PDF/DOC) so that only supported files are processed.

US4: Extract Resume Data
As an AI system, I want to extract key information from resumes so that candidate data can be structured.

US5: Categorize Candidate Information
As an AI system, I want to categorize skills, education, and experience so that candidates can be compared fairly.

US6: Flag Incomplete or Inaccurate Resumes
As an AI system, I want to flag incomplete or inconsistent resumes so that recruiters can follow up.

US7: View Structured Candidate Profiles
As a recruiter, I want to view structured candidate profiles so that reviewing applications is faster.

US8: Search & Filter Candidates
As a recruiter, I want to search and filter candidates based on extracted data so that I can find suitable candidates efficiently.

US9: Post Job Vacancy
As a recruiter, I want to create and post job vacancies with requirements so that I can attract suitable candidates.

US10: Learn from Historical Data
As an AI system, I want to learn from past successful candidates so that I can improve candidate evaluation.

Use Cases

Use Case 1: Upload Resume

Actor: Candidate
Steps:

1. Candidate opens the company portal.
2. Candidate selects a job vacancy.
3. Candidate uploads resume.
4. AI System validates the file format.
5. AI System stores the resume.
6. AI System sends confirmation email.

Use Case 2: Extract and Process Resume

Actor: AI System
Steps:

1. AI System receives uploaded resumes.
2. AI System extract key information (skills, education, and experience).
3. AI System categorizes extracted data.
4. AI System checks for missing or inconsistent data.
5. AI System flags issues if found.
6. AI System generates structured candidate profile.

Use Case 3: Review Candidates

Actor: Recruiter
Steps:

1. Recruiter logs into the system.
2. Recruiter views candidate profiles.
3. Recruiter applies filters/search criteria.

4. AI System displays matching candidates.

Use Case 4: Post Job Vacancy

Actor: Recruiter
Steps:

1. Recruiter opens job creation page.
2. Recruiter enters job description and requirements.
3. Recruiter submits the vacancy.
4. System publishes the job post.

Use Case 5: AI Learning from Historical Data

Actor: AI System
Steps:

1. AI System collects historical candidate data.
2. AI System identifies successful candidate patterns.
3. AI System updates evaluation model.
4. AI System improves candidate ranking accuracy.

Test Cases

Test Case	Input	Expected Result
TC1	Upload unsupported file (e.g., .jpg)	Validation error shown
TC2	Upload empty resume	Error message displayed
TC3	Upload valid PDF/DOC resume	Resume successfully uploaded
TC4	Submit an application.	Confirmation email received
TC5	The resume is missing key data	Resume flagged by AI System
TC6	Valid resume	Structured profile generated
TC7	Search with filters	Relevant candidates displayed
TC8	No matching candidates	Empty result message shown
TC9	Create job without required fields	Validation error shown

TC10	Valid job posting	Job successfully published and show the message " job vacancy posted."

BPMN 1:

1. Resume Upload Flow

● Start → Open Company Portal → Select Job Vacancy → Upload Resume → Submit Application → Validate Resume Format
● If Invalid → Show Error Message → Back to Upload Resume
● If Valid → Store Resume → Send Confirmation Email → End

2. Resume Processing Flow

● Start → Receive Uploaded Resume → Extract Key Information → Categorize Data → Check Data Completeness
● If Incomplete/Inconsistent → Flag Resume → Store with Warning → End
● If Complete → Generate Structured Candidate Profile → Store Profile → End

3. Recruiter Review Flow

● Start → Recruiter Login → View Candidate Profiles → Apply Search/Filter → Matching Candidates Found?
● If No → Show Message "No Results Found" → End
● If Yes → Display Matching Candidates → End

4. Job Posting Flow

● Start → Open Job Creation Page → Enter Job Details → Submit Job → Validate Required Fields

- If Invalid → Show Error Message → Back to Form
- If Valid → Posting Job Vacancy → Show “Job vacancy posted” Message → End

5. AI Learning Flow

- Start → Collect Historical Candidate Data → Analyze Successful Candidates → Update AI Model → Improve Ranking Accuracy → End

Business Process 2:

User Stories

US1: Evaluate Candidate
As a candidate, I want to be evaluated based on my qualifications so that I am assessed fairly.

US2: Score Candidates
As an AI system, I want to score candidates based on education, skills, experience, and courses so that recruiters receive a list of suitable candidates.

US3: Rank Candidates
As an AI system, I want to rank candidates based on their scores so that top candidates are prioritized.

US4: Learn from Recruiter Feedback
As an AI system, I want to learn from recruiter feedback so that scoring accuracy improves over time.

US5: View AI-generated Scores
As a recruiter, I want to view AI-generated candidate scores so that I can make better hiring decisions.

US6: Edit Candidate Scores
As a recruiter, I want to update AI-generated scores when needed so that human judgment is incorporated.

US7: Compare Candidates
As a recruiter, I want to compare closely ranked candidates so that I can shortlist efficiently.

US8: Match Candidates to Job Requirements
As an AI system, I want to match candidates' resumes with job requirements so that only relevant candidates are considered.

Use Cases

Use Case 1: Score Candidates

Actor: AI System
Steps:

1. AI system retrieves structured candidate data.
2. AI system evaluates education, skills, experience, certifications, projects, and courses.
3. An AI system assigns a score to each candidate.
4. AI system stores scores in the database.

Use Case 2: Rank Candidates

Actor: AI System
Steps:

1. AI system retrieves all candidate scores.
2. AI system sorts candidates based on scores.
3. AI system ranks candidates from highest to lowest.
4. AI system updates ranking list.

Use Case 3: Match Candidates to Job Requirements

Actor: AI System
Steps:

1. AI System retrieves job requirements.
2. AI System compares candidate data with job requirements.
3. AI System calculates matching scores.
4. AI System filters out unqualified candidates.

Use Case 4: Review Candidate Scores

Actor: Recruiter
Steps:

1. Recruiter logs into the system.
2. Recruiter views ranked candidate list.
3. Recruiter selects a candidate.
4. AI System displays detailed information about selected candidate.

Use Case 5: Adjust Candidate Scores

Actor: Recruiter
Steps:

1. Recruiter selects a candidate.
2. Recruiter edits the AI-generated score.
3. Recruiter submits the updated score.
4. AI System saves changes, updates ranking and improves scoring model.

Use Case 6: Compare Candidates

Actor: Recruiter
Steps:

1. Recruiter selects multiple candidates.
2. AI System displays head-to-head comparison.
3. Recruiter reviews differences in qualifications and scores.
4. Recruiter shortlists candidates.

Use Case 7: Learn from the recruiter's feedback

Actor: AI System
Steps:

1. AI system collects recruiter adjustments and feedback.
2. AI systems analyze patterns in adjustments.
3. System updates scoring model.
4. Future scoring reflects improved accuracy

Test Cases

Test Case	Input	Expected Result
TC1	Candidate with complete data	Score successfully generated
TC2	Candidate missing key data	Flag candidate regarding missing data
TC3	Multiple candidates scored	Ranking generated correctly
TC4	Candidates with same score	Same ranks handled correctly
TC5	Recruiter views scores	Scores displayed clearly
TC6	Recruiter updates the score.	Score updated and ranking refreshed
TC7	Recruiter compares candidates	Comparison view displayed
TC8	Poor match with job requirements	Candidate flagged as disqualified
TC9	Strong match with job requirements	Candidate ranked higher
TC10	Recruiter feedback applied	Future scoring improvements

BPMN 2:

1. Candidate Scoring Flow

● Start → Retrieve Structured Candidate Data → Evaluate (Education, Skills, Experience, Courses) → Assign Score → Store Score → End

2. Candidate Ranking Flow

● Start → Retrieve Candidate Scores → Sort Candidates → Rank from Highest to Lowest → Update Ranking List → End

3. Match Candidates to Job Requirements

● Start → Retrieve Job Requirements → Compare with Candidate Data → Calculate Matching Score → [Meets Requirements?]
● If No → Flag as Disqualified → End
● If Yes → Keep Candidate → End

4. Recruiter Review Scores Flow

● Start → Recruiter Login → View Ranked Candidate List → Select Candidate → Display Candidate Details & Score → End

5. Adjust Candidate Scores (Recruiter and AI)

● Start → Select Candidate → Edit AI Score → Submit Updated Score → Save Changes → Update Ranking → Improve AI Model → End

6. Compare Candidates Flow

● Start → Select Multiple Candidates → Generate Comparison View → Display Differences (Skills, Experience, Scores) → Recruiter Shortlists Candidates → End

7. AI Learning from Recruiter's feedback Flow

● Start → Collect Recruiter Feedback → Analyze Adjustments → Update Scoring Model → Improve Future Scoring Accuracy → End

Business Process 3:

User Stories

US1: Notify Shortlisted Candidates
As a candidate, I want to be notified when I am shortlisted so that I can prepare for the next step in the hiring process.

US2: Provide Feedback to Rejected Candidates
As a candidate, I want to receive feedback when I am not shortlisted so that I can improve future applications.

US3: Generate Candidates Shortlist
As an AI system, I want to generate a shortlist of qualified candidates based on scores so that recruiters can focus on the best matches.

US4: Provide Feedback for Decisions
As an AI system, I want to generate feedback for shortlisted and rejected candidates so that hiring decisions are transparent.

US5: Review Shortlisted Candidates
As a recruiter, I want to review AI-generated shortlists so that I can confirm or modify selected candidates.

US6: Contact Shortlisted Candidates
As a recruiter, I want to communicate with shortlisted candidates so that they can proceed to the next hiring stage.

US7: Archive or Reject Candidates

As a recruiter, I want to archive or reject non-shortlisted candidates so that the database remains organized and reusable.

US8: Match Candidates to Job Requirements
As an AI system, I want to match candidates with job requirements during shortlisting so that only suitable candidates are selected.

Use Cases

Use Case 1: Generate Shortlist

Actor: AI System
Steps:

1. AI System retrieves ranked candidates.
2. AI System applies job requirement filters.
3. AI System selects top candidates.
4. AI System generates shortlists.
5. AI System attaches feedback for selection.

Use Case 2: Notify Candidates

Actor: AI System
Steps:

1. AI System identifies shortlisted candidates.
2. AI System sends notification shortlisted candidates through email.
3. System updates candidate's hiring status.

Use Case 3: Provide Feedback to Rejected Candidates

Actor: AI System
Steps:

1. AI System identifies non-shortlisted candidates.
2. AI System generates feedback based on missing data or requirements.
3. AI System sends feedback notification.

Use Case 4: Review Shortlist

Actor: Recruiter
Steps:

1. Recruiter logs into the system.
2. Recruiter views AI-generated shortlist.
3. Recruiter reviews candidate details.
4. Recruiter confirms or modifies shortlist.

Use Case 5: Communicate with Candidates

Actor: Recruiter
Steps:

1. Recruiter selects shortlisted candidates.
2. The recruiter contacts selected candidates regarding the interview or the following step in the hiring process.
3. AI systems access communication history.

Use Case 6: Archive/Reject Candidates

Actor: Recruiter
Steps:

1. Recruiter selects non-shortlisted candidates.
2. Recruiter marks candidates as rejected or archived.
3. AI System stores candidates in a centralized database for future reference.

Test Cases

Test Case	Input	Expected Result
TC1	Candidates with high scores	Added in shortlist
TC2	Candidates below threshold	Not shortlisted
TC3	Shortlist generated	Only top candidates displayed
TC4	Candidate shortlisted	Notification sent
TC5	Candidate rejected	Sends rejection email with constructive feedback
TC6	Recruiter modifies shortlist	Changes saved successfully
TC7	Recruiter sends message	Candidate receives email message
TC8	Recruiter archives candidate	Candidate stored for future reference
TC9	Matching with job requirements	Only relevant candidates shortlisted
TC10	No suitable candidates	Empty shortlist message shown quoting “ No suitable candidates”

BPMN 3:

1. Generate Shortlist (AI)

● Start → Retrieve Ranked Candidates → Apply Job Requirement Filters → Select Top Candidates → Generate Shortlist → Attach Feedback → End

2. Notify Shortlisted Candidates (AI)

● Start → Identify Shortlisted Candidates → Send Notification Email → Update Candidate Status → End

3. Provide Feedback to Rejected Candidates (AI)

● Start → Identify Non-Shortlisted Candidates → Generate Feedback → Send Rejection Email → End

4. Recruiter Review Shortlist

● Start → Recruiter Login → View AI Shortlist → Review Candidate Details → [Modify Shortlist?]
● If Yes → Update Shortlist → Save Changes → End
● If No → Confirm Shortlist → End

5. Communicate with Candidates

● Start → Select Shortlisted Candidates → Send Message (Interview / Next Step) → Log Communication History → End

6. Archive / Reject Candidates

● Start → Select Non-Shortlisted Candidates → Mark as Rejected or Archived → Store in Database → End

Template Prompt

1. Context

You are helping an HR department in a large company implementing an AI-powered recruitment system.
The company currently faces challenges in manual resume screening, including time-consuming processes, inconsistent evaluations, and potential human bias.

As mentioned in the research, traditional recruitment methods do not align with scalability, fairness, and efficiency, especially when managing large volumes of applications. The company aims to adopt an automated applicant screening system using artificial intelligence to improve hiring decisions while maintaining fairness and transparency.

2. Task

Design a complete automated applicant screening system using artificial intelligence that supports:

Resume collection and data extraction, candidate scoring and ranking. Also, candidate shortlisting while maintaining communication

The system should integrate AI power such as resume parsing, job matching and predictive evaluation.

3. Requirements

Resume Collection & Data Extraction

- Candidates can upload resumes through a portal
- System validates file formats (PDF/DOC)
- AI extracts key data (skills, education, experience)
- System categorizes candidate profiles
- AI flags incompleteness or inaccuracies occurred in resumes
- Recruiters can search and filter candidates
- AI System sends confirmation emails

Scoring & Ranking

- AI scores candidates based on:

  - Skills

  - Education

  - Experience

  - Certifications/courses

  - Projects

- AI ranks candidates based on scores
- AI matches resumes with job requirements using similarity techniques such as TF-IDF or Cosine Similarity.
- System learns from recruiter feedback (historical data) to improve scoring
- Recruiters can edit AI-generated scores
- Recruiters can compare candidates

Shortlisting

- AI generates shortlist of top candidates
- Candidates receive notifications if shortlisted
- Rejected candidates receive feedback
- Recruiters can review and modify shortlist
- Recruiters can communicate with shortlisted candidates
- System allows archiving/rejecting candidates

4. Constraints

- Must comply with Egyptian Labor Law No. 12 of 2003 (no discrimination)
- Must comply with Egypt Personal Data Protection Law No. 151 of 2020
- Must avoid bias in AI decision-making
- Must exclude sensitive personal attributes from scoring
- Must include human-in-the-loop (HR final decision)
- Must provide explainable scoring results

5. Output Format

Generate the system specification in the following structure:

Section 1: System Overview

- Purpose
- Scope
- Objectives

Section 2: User Roles

- Candidate
- Recruiter
- AI System

Section 3: Functional Requirements

- Organized by modules:
    - Resume Collection
    - Scoring & Ranking
    - Shortlisting

Section 4: Use Cases

- Detailed use cases with actors, steps and bpmns.

Section 5: Data Processing & AI Logic

- Resume parsing
- Feature/ key information extraction
- Matching models (TF-IDF, similarity models)
- Scoring logic

Section 6: Non-Functional Requirements

- Performance
- Scalability
- Security
- Fairness & Ethics

Section 7: Challenges & Limitations

- Bias
- Data quality
- Legal concerns